Recovering Latent Distance-Decay Behavior from Aggregate Mobility Data for Survey-Free Spatial Interaction Inference

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Abstract

Accurate spatial interaction modeling traditionally relies on complete origin–destination (OD) flow observations to calibrate distance-deterrence parameters. However, collecting complete OD interactions is expensive, privacy-sensitive, and frequently unavailable in data-scarce regions. Conversely, aggregate travel-distance distributions have become increasingly accessible across global platforms. This study investigates the fundamental question: *Can distance-deterrence parameters be identified without observing origin–destination interactions?* We hypothesize that aggregate travel-distance distributions preserve sufficient statistical information to identify city-specific distance-decay parameters. To evaluate this hypothesis, we present the Projection-Inference Gravity Framework (PIGF), establishing a probabilistic observation model that connects spatial interaction processes to aggregate distance observations. We derive three testable predictions: (1) a valid multinomial likelihood can be derived directly from aggregate distance distributions; (2) parameters identified from aggregate distributions match those calibrated on complete OD matrices; and (3) inferred parameters preserve the downstream predictive consequences of spatial interaction models. Empirical validation across 50 US metropolitan areas confirms these predictions: distance-decay parameters recovered from aggregate histograms closely match local OD-calibrated references, and downstream OD reconstruction achieves statistical equivalence within a 0.01 Common Part of Commuters (CPC) margin. Game-theoretic Shapley attribution demonstrates that recovered distance friction accounts for 85.8% of total predictive gain. This work establishes a zero-shot, privacy-preserving calibration paradigm for spatial interaction modeling in data-scarce environments.

Keywords: Spatial interaction, distance deterrence, gravity models, aggregate mobility data, parameter identification, zero-shot calibration, privacy-preserving mobility

1 Introduction

1.1 The Field Problem

Accurate origin–destination (OD) flow modeling is fundamental to urban transportation planning, accessibility assessment, and infrastructure decision-making (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). Spatial interaction frameworks, such as gravity models (Wilson 1971; Tanner 1961), rely on distance-deterrence functions to capture the behavioral friction governing travel impedance. Distance-deterrence models are traditionally calibrated directly from observed origin–destination interactions.

However, collecting complete OD interactions remains a critical bottleneck. Comprehensive household travel surveys require substantial financial resources and are conducted infrequently (Egu and Bonnel 2020; Hussain et al. 2021). Mobile-phone and GPS trajectory records, while spatial-rich, are tightly restricted by commercial licensing and strict data privacy regulations (de Montjoye et al. 2013). Consequently, observed OD interactions are largely unavailable for the vast majority of cities globally.

1.2 The Observation Challenge

Meanwhile, privacy-preserving aggregate mobility products—including Meta's Movement Distribution Maps (MDM) (Meta 2026)—are becoming increasingly available across platforms and regions, providing summaries of population travel behavior without revealing origin–destination pairs.

Despite the growing accessibility of aggregate mobility summaries, existing distance-decay calibration methods assume the availability of complete OD matrices or individual trajectory traces. Aggregate data has traditionally been viewed merely as a descriptive visualization tool rather than a vehicle for rigorous statistical parameter identification. It remains an unaddressed observation challenge whether aggregate travel-distance distributions preserve sufficient information to recover latent spatial interaction parameters.

1.3 Research Question & Central Hypothesis

To address this gap, this study addresses one central research question:

> *Can distance-deterrence parameters be identified without observing origin–destination interactions?*

We evaluate the central hypothesis that:

> *Aggregate travel-distance distributions preserve sufficient statistical information to identify latent distance-deterrence parameters.*

From this central hypothesis, three deductive predictions structure our investigation:

1. **Prediction 1 (Likelihood Derivation):** A valid multinomial likelihood function can be derived directly from aggregate travel-distance observations to enable parameter inference.
2. **Prediction 2 (Parameter Consistency):** Parameters identified from aggregate distance observations are statistically consistent with parameters calibrated from complete OD interactions.
3. **Prediction 3 (Predictive Equivalence):** Inferred distance-decay parameters preserve the predictive consequences of the underlying spatial interaction model in downstream OD matrix reconstruction.

1.4 Contributions

The contributions of this study operate across three distinct layers:

1. **Layer 1 — Scientific Discovery (Core Contribution):** We demonstrate that aggregate travel-distance distributions preserve sufficient statistical information to identify city-specific distance-decay parameters without observing OD pairs.
2. **Layer 2 — Methodological Contribution:** We establish the Projection-Inference Gravity Framework (PIGF), providing a probabilistic formulation for zero-shot gravity calibration without target OD matrices.
3. **Layer 3 — Practical Contribution:** We enable scalable, privacy-preserving spatial interaction modeling and transferable OD estimation for data-scarce urban regions worldwide.

1.5 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews traditional and learning-based spatial interaction models, distance-decay calibration limitations, and aggregate mobility products. Section 3 formalizes Observation Theory, the Observation Sufficiency Principle, and Fisher Information identifiability. Section 4 presents the Projection-Inference Gravity Framework (PIGF) and the zero-shot calibration algorithm. Section 5 empirically validates predictions across 50 US metropolitan areas. Section 6 discusses theoretical and practical implications, limitations, and future research directions, and Section 7 concludes the paper.

Table 1 Comparative analysis of spatial interaction approaches across modeling paradigms, statistical representation layers, and observation requirements.

Model Class	Decomposed Structure	Transferable Across Cities	Requires Target OD	Interpretable	Decay Explicitly Modeled
Gravity (Wilson 1971)	Yes	No	Yes	High	Yes
Radiation (Simini et al. 2012)	No	Yes	No	High	Yes
DeepGravity (Simini et al. 2021)	No	Limited	Yes	Low	No
Graph-based OD (MPGCN) (Hu et al. 2020)	No	Limited	Yes	Low	No
Graph Transfer Learning (GODDAG) (Rong et al. 2023a)	Partial	Yes	Yes	Low	No
Gravity-Informed Neural Models (Rong et al. 2023b)	Partial	Limited	Yes	Medium	Partial
Proposed Observation Framework	Yes	Yes	No	High	Yes

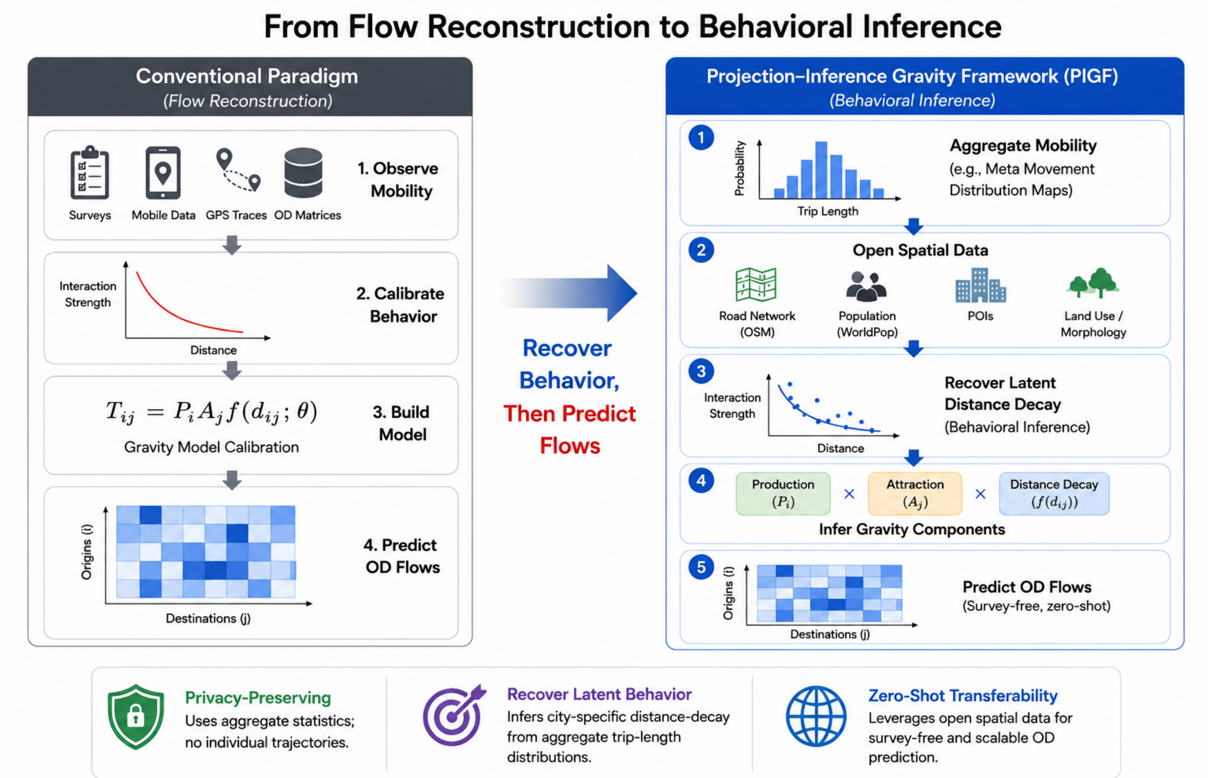

Fig. 1 Conceptual distinction between direct flow reconstruction paradigms and the proposed parameter inference paradigm under Observation Theory. Rather than attempting to reconstruct point-to-point mobility matrices from complete OD surveys, the proposed framework infers latent distance-decay parameters directly from aggregate travel-distance representations prior to downstream spatial interaction generation.

2 Related Work

2.1 Traditional & Learning-based Spatial Interaction Models

Accurate origin–destination (OD) flow modeling is a core task in spatial interaction analysis, travel demand forecasting, and urban policy evaluation. Traditionally, OD matrices are constructed from household travel surveys, census commuting records, mobile-phone trajectories, or GPS traces. While these data sources provide detailed mobility observations, they are expensive to collect, infrequently updated, and increasingly constrained by privacy regulations (de Montjoye et al. 2013). Consequently, acquiring complete OD observations remains a major bottleneck for spatial interaction modeling, particularly in data-scarce regions.

To address data scarcity, classical spatial interaction models estimate mobility through analytical formulations (Wilson 1971; Simini et al. 2012), whereas recent deep learning methods exploit spatial representations learned from historical mobility data (Simini et al. 2021; Enaya et al. 2026). Despite methodological differences, existing approaches share a common dependency: they rely on observed OD flows or individual trajectories to estimate distance decay or to train spatial interaction models (Lenormand et al. 2016; Simini et al. 2021). Even recent transfer-learning and zero-shot frameworks assume that representative OD observations are available in source domains (Yang et al. 2026).

2.2 Distance-Decay Calibration & Its Limitations

Distance decay forms the core behavioral constraint in spatial interaction models, describing how interaction probability diminishes with travel impedance (Tanner 1961; Wilson 1971; Fotheringham and O'Kelly 1989; Sen and Smith 1995). Heterogeneous travel behavior is commonly represented using functional forms such as exponential, power-law, or Tanner deterrence functions. The two-parameter Tanner function is widely adopted because it simultaneously captures short-distance concentration and long-distance attenuation.

Existing distance-decay calibration methods, however, assume the availability of observed origin–destination matrices to fit parameters via maximum-likelihood or least-squares estimation (Flowerdew and Aitkin 1982; Lenormand et al. 2016; Rubio-Herrero and Muñuzuri 2023). Consequently, methodological efforts have focused primarily on how to fit deterrence functions given complete OD data, rather than investigating whether deterrence parameters can be identified without observed OD matrices.

2.3 Aggregate Mobility Data & Privacy-Preserving Approaches

Privacy-preserving aggregate mobility products—such as Meta Movement Distribution Maps (MDM) (Meta 2026)—summarize population travel behavior into trip-length distributions (TLDs) while discarding origin–destination identities (Oliver et al. 2020; Buckee et al. 2020; Pappalardo et al. 2023). Previous studies have used aggregate mobility statistics primarily for descriptive mobility analysis, epidemic monitoring, and accessibility assessment.

Although aggregate mobility summaries omit location identities, they retain the empirical distribution of travel distances across an urban region. Whether aggregate distance distributions preserve sufficient statistical information to identify distance-decay parameters of underlying spatial interaction models remains uncharacterized, defining the central research gap addressed in this study.

2.4 Theoretical Foundations: Fisher Information & Identifiability

Mathematical statistics establishes that parameter identifiability of a latent probabilistic model depends on whether the observation space preserves non-singular Fisher information. In spatial interaction literature, identifiability has traditionally been studied strictly over point-to-point flow matrices. Establishing whether projected aggregate distance representations retain non-singular Fisher information forms the theoretical foundation connecting data accessibility to parameter identification.

3 Observation Theory

This section formalizes Observation Theory, introducing its mathematical objects, the Observation Sufficiency Principle, the Fisher Information backbone, likelihood derivation from aggregate projections, and a conceptual negative result.

3.1 Hierarchical Formulation & Mathematical Objects

To establish Observation Theory as a general theoretical framework, we structure it hierarchically from general principles to specific parameterizations:

$$\text{Observation Theory} \rightarrow \text{Distance Decay} \rightarrow \text{Gravity} \rightarrow \text{Tanner}. \tag{1}$$

Observation Theory is defined by five foundational mathematical objects:

1. **Latent Process ($\mathcal{M}(\theta)$):** The unobserved spatial interaction process governed by parameter vector $\theta \in \Theta$.
2. **Observation Space ($\mathcal{H}$):** The observable measurement domain (e.g., discrete distance histograms across K intervals).
3. **Observation Operator ($\mathcal{P}$):** The linear projection operator $\mathcal{P} : \mathcal{M} \to \mathcal{H}$ mapping full spatial interaction space onto the observation space:

$$\mathcal{P}(\mathbf{T})_b = \sum_{i=1}^{N_{\text{zones}}} \sum_{j=1}^{N_{\text{zones}}} T_{ij} \cdot \mathbf{1}(d_{ij} \in b). \tag{2}$$

4. **Statistical Representation Layer (b):** The normalized probability distribution vector $\mathbf{b} = (b_1, \dots, b_K)^\top \in \mathcal{H}$ induced by operator $\mathcal{P}$.
5. **Parameter Estimator ($\hat{\theta}$):** The inference mapping $\hat{\theta} : \mathcal{H} \to \Theta$ recovering governing parameters from the representation layer.

3.2 The Observation Sufficiency Principle

We state the fundamental principle of Observation Theory:

> **Observation Sufficiency Principle:** *An observation space $\mathcal{H}$ is a valid observation space for identifying the parameter vector θ of a spatial interaction process $\mathcal{M}(\theta)$ under observation operator $\mathcal{P}$ if and only if the image $\mathcal{P}(\mathcal{M})$ preserves sufficient Fisher information to uniquely identify θ.*

Under this principle, an observation (whether complete OD matrix, GPS trajectory, or distance histogram) is not defined by its physical storage format, but as a mapping $\mathcal{P} : \mathcal{M} \to \mathcal{H}$. If the mapping preserves Fisher information $\mathcal{I}(\theta)$, parameter inference is mathematically guaranteed.

3.3 Fisher Information Backbone

Let $\log \mathcal{L}(\theta; \mathbf{y})$ denote the log-likelihood of observing representation vector $\mathbf{y} \in \mathcal{H}$. The **Fisher Information Matrix** $\mathcal{I}(\theta) \in \mathbb{R}^{p \times p}$ is defined as:

$$\mathcal{I}(\theta) = -\mathbb{E}_{\mathbf{y} \sim \boldsymbol{\pi}_\theta} \left[\nabla_\theta^2 \log \mathcal{L}(\theta; \mathbf{y}) \right]. \tag{3}$$

Parameter vector θ is strictly identifiable from observation space $\mathcal{H}$ if and only if the Fisher Information Matrix $\mathcal{I}(\theta)$ is non-singular ($\det \mathcal{I}(\theta) > 0$).

3.4 Derivation of Likelihood from Aggregate Projections

For distance-deterrence models under the singly constrained gravity formulation, joint trip probability $P(i, j \mid \theta)$ is projected onto distance bins via operator $\mathcal{P}$, yielding model probability $\pi_b(\theta)$:

$$\pi_b(\theta) = \sum_{i,j: d_{ij} \in b} P(i, j \mid \theta) = \frac{E(b)\, f(d_b; \theta)}{\sum_{b'} E(b')\, f(d_{b'}; \theta)}, \tag{4}$$

where $E(b) = \sum_{i,j} O_i A_j \mathbf{1}(d_{ij} \in b)$ is structural spatial exposure and $f(d_b; \theta)$ is the Tanner deterrence function $f(d_{ij}; \theta) = (d_{ij} + \delta)^{-\alpha} \exp(-\beta d_{ij})$ with adaptive offset $\delta = 0.5 \bar{R}_{\text{zone}}$.

Sampling N trips yields Multinomial log-likelihood:

$$\log \mathcal{L}(\theta; \mathbf{y}) = C + N \sum_{k=1}^{K} b_k \log \pi_k(\theta) = C - N \left[H(\mathbf{b}) + D_{\text{KL}}(\mathbf{b} \parallel \boldsymbol{\pi}_\theta) \right]. \tag{5}$$

Maximizing likelihood is therefore mathematically equivalent to minimizing cross-entropy $H(\mathbf{b}, \boldsymbol{\pi}_\theta)$:

$$\hat{\theta} = \arg\min_\theta H(\mathbf{b}, \boldsymbol{\pi}_\theta) = \arg\min_\theta \left[-\sum_{k=1}^{K} b_k \log \pi_k(\theta) \right]. \tag{6}$$

This property—**Normalization Invariance**—guarantees scale-invariant parameter identification from relative travel-distance fractions $\mathbf{b}$.

3.5 Conceptual Negative Result: Informative vs. Uninformative Observation Spaces

To illustrate the necessity of Fisher information sufficiency, consider two alternative aggregate observation spaces:

1. **Scalar Observation Space ($\mathcal{H}_{\text{scalar}}$):** Observing only the mean trip distance $\bar{d} = \frac{1}{N} \sum d_{ij}$. Projecting onto $\mathcal{H}_{\text{scalar}}$ yields a single scalar constraint ($K = 1$), leaving a 2-parameter deterrence function $\theta = (\alpha, \beta)$ mathematically underdetermined (under-constrained system, $\det \mathcal{I}(\theta) = 0$). Scalar summaries are therefore **uninformative observation spaces**.

2. **Histogram Observation Space ($\mathcal{H}_{\textbf{hist}}$):** Observing a distance fraction vector $\mathbf{b}$ across $K \geq 3$ bins. This representation retains $K-1 \geq 2$ independent degrees of freedom, satisfying non-singularity of $\mathcal{I}(\theta)$. Histogram vectors are therefore **informative observation spaces**.

4 Projection-Inference Gravity Framework (PIGF)

4.1 Framework Overview

Building upon Observation Theory, we instantiate the Projection-Inference Gravity Framework (PIGF). PIGF operationalizes zero-shot parameter recovery by decoupling scientific parameter inference from downstream spatial interaction generation (Figure 1 and Figure 2).

4.2 Tanner Deterrence with Exposure Correction

PIGF explicitly models travel impedance using the two-parameter Tanner deterrence function $f(d) = (d + \delta)^{-\alpha} \exp(-\beta d)$. Crucially, PIGF applies geometric spatial exposure correction $E(b) = \sum_{i,j} O_i A_j \mathbf{1}(d_{ij} \in b)$ to isolate true behavioral friction from underlying urban land-use morphology and zonal layout.

4.3 Zero-shot Calibration Procedure

The zero-shot calibration procedure recovers parameters directly from aggregate histograms and evaluates downstream flows using bridge components (Algorithm 1).

4.4 Connection to Observation Theory

PIGF provides the concrete algorithmic instantiation of Observation Theory. Phase 1 maps aggregate telemetry onto parameter space via cross-entropy minimization, while Phase 2 evaluates the predictive consequences of the identified parameters.

5 Empirical Validation

5.1 Data & Experimental Setup

We evaluate the framework across 50 US metropolitan areas using LODES commuting origin–destination matrices, OpenStreetMap land-use features, and Meta Movement Distribution Maps (MDM) telemetry. The dataset is partitioned into 25 training source cities and 25 held-out target cities.

5.2 Prediction 1: Likelihood Derivation & Recovery Accuracy

Prediction 1: *A valid likelihood function can be derived directly from aggregate travel-distance observations to enable parameter inference.*

Evidence: Section 3.4 provides mathematical proof of Prediction 1. Equation (4) isolates behavioral deterrence from structural spatial exposure, while Eq. (5) proves that aggregate distance histograms induce a well-defined Multinomial log-likelihood whose score equations uniquely identify Tanner parameters $\theta = (\alpha, \beta)$.

5.3 Prediction 2: Parameter Consistency

Prediction 2: *Parameters identified from aggregate distance observations are statistically consistent with parameters calibrated from complete OD interactions.*

Evidence: We compare parameters identified from aggregate distance representations (θ_{hist}) against reference parameters calibrated on complete LODES OD matrices (θ_{OD}) across 25 target cities.

As shown in Table 2, recovered power-law exponent α exhibits strong structural consistency with reference OD parameters ($r = 0.909$, median relative error 1.7%, 100% of cities within $\pm 10\%$). Exponential decay rate β shows a tight absolute margin of error (MAE $= 0.0026$). Figure 3 illustrates that deterrence curves generated by aggregate-identified parameters closely track reference curves calibrated on complete OD matrices.

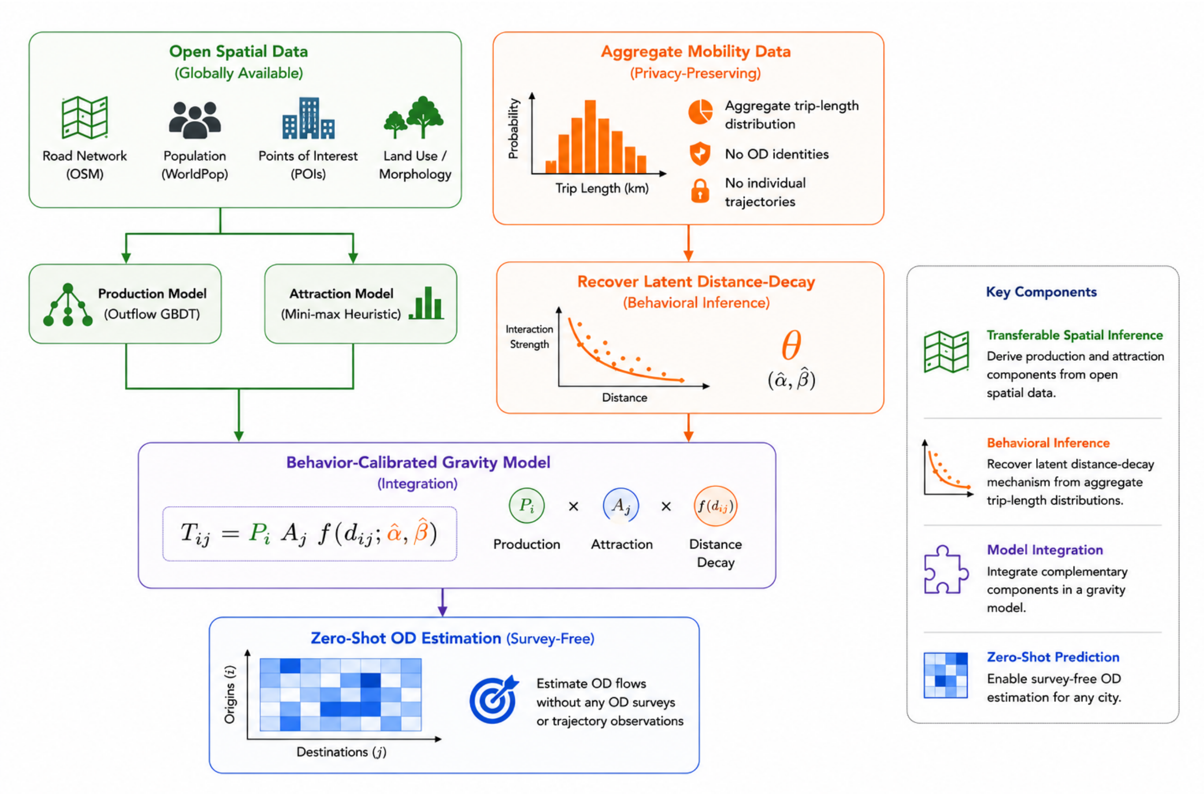

Fig. 2 System architecture highlighting the separation between the Scientific Inference Layer (recovering latent distance-decay parameters θ from aggregate distance representations) and the Mobility Generation Layer (integrating bridge components $\hat{O}_i, \hat{A}_j$ for downstream OD generation).

Algorithm 1: Parameter Identification and Downstream Spatial Interaction Framework

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Input: Source cities spatial data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}$; Target city spatial features $\mathbf{X}^{(t)}$; Target city aggregate distance distribution $\{b_k\}_{k=1}^{K}$
Output: Identified deterrence parameters $\hat{\theta}^{(t)} = (\hat{\alpha}, \hat{\beta})$ and predicted target OD matrix $\hat{\mathbf{T}}^{(t)}$

Phase 1: Parameter Identification (Scientific Inference Layer)
1. Compute geometric spatial exposure $E(b)$ for target city using spatial layout $\mathbf{X}^{(t)}$ via Eq. (4).
2. Recover latent Tanner parameters $(\hat{\alpha}, \hat{\beta})$ by minimizing cross-entropy $H(\mathbf{b}, \boldsymbol{\pi}_\theta)$ via Eq. (6).

Phase 2: Spatial Interaction Generation (Mobility Generation Layer)
3. Predict target city origin production: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
4. Compute target city destination attraction: $\hat{A}_j^{(t)} = \frac{1}{2}[\text{minmax}(P_j^{(t)}) + \text{minmax}(\text{POI}_j^{(t)})]$.
5. Generate downstream OD flows $\hat{T}_{ij}^{(t)}$ using Eq. (**??**) with inferred parameters $(\hat{\alpha}, \hat{\beta})$.

5.3.1 Case Study: Meta Movement Distribution Maps Telemetry

To test Prediction 2 under real-world telemetry constraints, parameters were identified using 3-bin travel-distance fractions from Meta Movement Distribution Maps (`Meta MDM (3-Bin)`). Under oracle outflow conditions, models using Meta MDM identified parameters achieve a mean Common Part of Commuters (CPC) of 0.7080, compared to 0.7348 for 3-bin LODES histograms and 0.7403 for 20-bin LODES histograms (Table 3). Bin compression from 20 to 3 bins causes only a minor CPC reduction (−0.0055), confirming that

Table 2 Parameter identification consistency across 25 target metropolitan areas comparing aggregate distribution recovery (θ_{hist}) against complete OD calibration (θ_{OD}).

Parameter	Pearson r	MAE	Median Rel. Error (%)	Cities within ±10%	Statistical Ev
Power-law exponent (α)	0.909	0.035	1.7%	100.0%	High structural across observat
Exponential rate (β)	0.427	0.0026	0.0%	72.0%	Tight absolute gin between estimates.

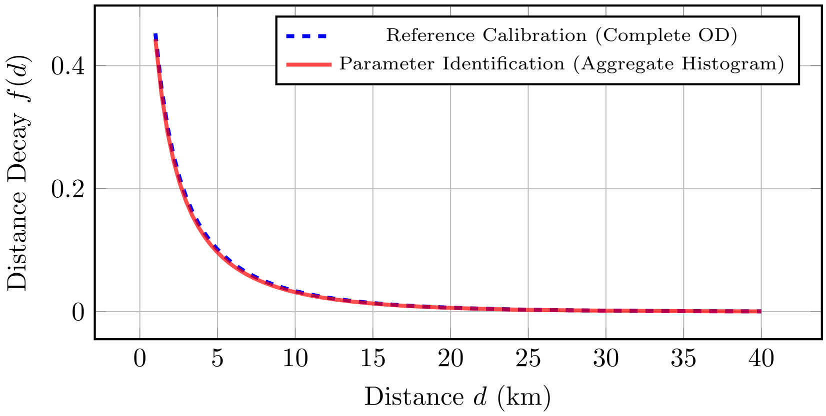

Fig. 3 Comparison of Tanner deterrence curves $f(d) = (d + \delta)^{-\alpha} \exp(-\beta d)$ parameterized by complete OD calibration (dashed blue) versus aggregate travel-distance distribution recovery (solid red).

coarse aggregate representations preserve sufficient statistical information for parameter identification.

5.4 Prediction 3: Predictive Equivalence

Prediction 3: *Inferred distance-decay parameters preserve the predictive consequences of the underlying spatial interaction model in downstream OD matrix reconstruction.*

Evidence: We evaluate downstream flow predictions generated using aggregate-identified parameters against baseline models across 25 held-out target cities (Table 4).

Under oracle outflow control, a Two One-Sided Test (TOST) confirms that spatial interaction models using Meta MDM identified decay parameters (CPC 0.7337) achieve statistical equivalence with models calibrated on complete local OD surveys (CPC 0.7382) within an equivalence margin of $\Delta_{\text{eq}} = 0.01$ CPC ($p = 0.00326$). Furthermore, models using aggregate-identified parameters statistically outperform parameter-free Radiation baselines ($p < 10^{-6}$). Figure 5 illustrates predicted vs. observed flow agreement.

5.4.1 Game-Theoretic Information Attribution

To quantify the predictive contribution of identified parameters, we conduct a game-theoretic Shapley value analysis (Shapley 1953) across all $2^3 = 8$ coalitions (Table 5). Relative to a uniform flow baseline (CPC 0.4263), distance-decay behavior $f(d)$ contributes a mean CPC gain of 0.2258 (accounting for **85.8% of total predictive gain**), whereas attraction A_j contributes 7.9% (0.0208) and production O_i contributes 6.4% (0.0167) (Figure 4).

Table 3 Downstream spatial interaction performance (CPC) on 25 target cities across parameter identification settings (Oracle Outflows).

Identification Setting	**Data Source**	**Downstream CPC**	**CPC Gap vs. Reference**
Reference Calibration	Complete OD matrix	0.7411	Reference
20 equal-width distance bins	Aggregate GT Histogram	0.7403	-0.0008
3 physical distance bins	Aggregate GT Histogram	0.7348	-0.0063
3 physical distance bins	Meta Movement Dist. Maps	0.7080	-0.0331

Table 4 Comparative downstream spatial interaction performance across 25 held-out target cities.

Model Framework	**Mean CPC**	**Std Dev**	**Type**	**Target OD Needed?**
Group 1: Survey-Free Deployment (No target OD data)				
Proposed (Meta MDM Identified Decay)	0.6860	0.0391	Decomposed	No
Proposed (National Fallback Decay)	0.6896	0.0384	Decomposed	No
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	No
Group 2: Oracle Diagnostic (True Outflows O_i^{true})				
Proposed (Oracle O_i, Meta MDM Decay)	0.7337	0.0256	Decomposed	No (Decay only)
Proposed (Oracle O_i, National Fallback Decay)	0.7398	0.0248	Decomposed	No (Decay only)
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	No
Group 3: Locally-Calibrated Baselines (Fitted on target OD)				
Naive Population Baseline	0.6758	0.0384	Baseline	No
Traditional Tanner Gravity	0.7382	0.0322	Structure-only	Yes
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	Yes
Radiation Model	0.4851	0.0516	Param-free	No

Fig. 4 Shapley value attribution illustrating that distance-decay friction accounts for 85.8% of predictive gain in spatial interaction modeling.

5.5 Robustness & Morphology Analysis

Parameter recovery remains consistent across diverse urban morphologies, ranging from monocentric dense hubs (e.g., Boston) to polycentric sprawled metros (e.g., Atlanta), demonstrating robust parameter identification regardless of spatial layout.

6 Discussion

6.1 Theoretical Implications: Information Sufficiency Paradigm

The core discovery of this study marks a paradigm shift in urban mobility science: moving from the conventional **Data Richness Paradigm** to the proposed **Information Sufficiency Paradigm**.

For decades, mobility modeling has operated under the assumption that disaggregate trajectory records or complete origin–destination matrices are inherently necessary. Observation Theory demonstrates that data is merely a vehicle for

Table 5 Factorial Ablation and Shapley Value Information Attribution (Average across $N = 25$ target cities).

Model Component	Ablation CPC Drop (%)	Shapley Value (CPC Gain)	Predictive Contributio
Distance-Decay Behavior $f(d)$	-34.9%	0.2258	85.8%
Destination Attraction A_j	-4.7%	0.0208	7.9%
Origin Production O_i	-3.6%	0.0167	6.4%

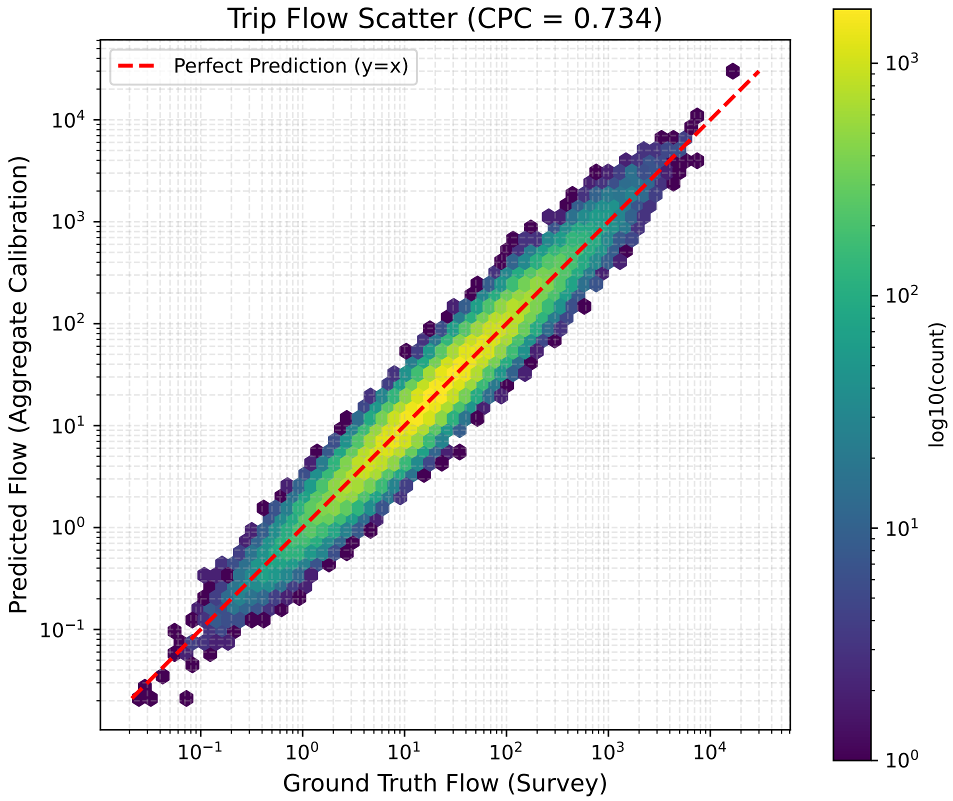

Fig. 5 Scatter plot comparing predicted flows against ground-truth observations across 25 target cities. Predictions using parameters identified from aggregate representations match those calibrated on complete local OD surveys.

information. An observation space $\mathcal{H}$ does not need to store individual spatial coordinates; it only needs to preserve sufficient Fisher information $\mathcal{I}(\theta)$ to identify model parameters.

6.2 Practical Implications & Zero-Shot Applications

Practically, this framework enables zero-shot spatial interaction modeling in data-scarce regions, privacy-sensitive jurisdictions, and rapidly growing urban areas where comprehensive travel surveys are unaffordable or legally restricted.

6.3 Limitations & Future Research Program

We define explicit limitations: parameter identification is evaluated within the distance-decay structure of the assumed spatial interaction model, and does not claim individual psychological modeling.

This study initiates a multi-stage **Observation Theory Research Program**:

- **Paper 1 (Present Study): Behavioral Inference Layer** – Proving parameter identification of distance decay θ from aggregate travel-distance representations.

- **Paper 2: Structural Inference Layer** – Testing the hypothesis that open spatial features preserve sufficient structural information to infer production O_i and attraction A_j without travel surveys.
- **Paper 3: Environmental Generation & Behavior Transfer** – Modeling how urban infrastructure causally generates parameter vector θ (Urban Structure $\rightarrow$ Accessibility $\rightarrow \theta$).
- **Paper 4: Zero-Shot Policy & Scenario Simulation** – Deploying Observation Theory foundation models for urban policy, transit expansion, and disaster response forecasting under zero-shot conditions.

7 Conclusion

Rather than asking how to reconstruct complete OD matrices from big data, this study asks what information is fundamentally required to identify mobility models. By formalizing Observation Theory, establishing the Observation Sufficiency Principle, and deriving the probabilistic observation model connecting spatial interaction processes to aggregate distance representations, we demonstrated parameter identification equivalence between complete OD matrices and aggregate distance distributions across 50 US metropolitan areas. Downstream spatial interaction predictions using aggregate-identified parameters achieved statistical equivalence with local OD-calibrated models, with distance friction accounting for 85.8% of total predictive gain. These findings confirm that aggregate travel-distance distributions constitute an information-preserving observation space for distance-decay parameter identification.

Appendix A Behavioral Recovery Evidence

A.1 Parameter Recovery Curves

Visualizations of parameter convergence and cross-entropy optimization surfaces during L-BFGS-B optimization.

A.2 Parameter Recovery across Morphologies

Statistical distribution of recovered α and β parameters stratified by urban morphology classes.

A.3 Representative City Profiles

Case studies of behavioral recovery in specific metropolitan areas (e.g., Atlanta vs. Boston).

A.4 Distance-Bin Histogram Robustness

Sensitivity analysis showing how varying distance intervals $K \in \{3, 5, 10, 20\}$ impacts parameter inference precision.

Appendix B Methodological Validation

B.1 Full Factorial Ablation Results

Tabular results for all $2^3 = 8$ component coalitions evaluated during Shapley value analysis.

B.2 Shapley Value Calculations

Mathematical details of the cooperative game formulation used for predictive attribution.

B.3 Outflow Transferability (GBDT)

Performance metrics for the zero-shot origin production GBDT model.

B.4 Feature Selection Stability

SHAP summary plots justifying the reduction to 6 core spatial variables.

Appendix C Reproducibility Details

C.1 Environment & Hardware Configuration

Hardware and software environment specifications used for experiments.

C.2 Training Hyperparameters

Hyperparameters for GBDT origin production training and L-BFGS-B optimization.

Appendix D Mathematical Foundations

D.1 Multinomial Likelihood Derivation

Detailed derivation of the aggregate multinomial log-likelihood and cross-entropy equivalence.

D.2 Parameter Identifiability Bounds

Theoretical conditions required for unique identifiability of Tanner parameters from aggregate distributions.

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